Terminal p.d. $= 4.0 - (0.80 \times 1.0) = 3.2$ V

24 **C** $BIL\sin\theta - mg = ma$

$0.040I\left(0.30\sin 90^\circ\right) - 0.0024(9.81) = 0.0024(1.2)$

$I = 2.202$ A

When $I = \dfrac{I}{2} = 1.101$ A

$BIL\sin\theta - mg = ma$

$0.040(1.101)\left(0.30\sin 90^\circ\right) - 0.0024(9.81) = 0.0024a$

$a = -4.305\ \text{m s}^{-2} = 4.3\ \text{m s}^{-2}$ (downwards)

25 **C** When wire Y is perpendicular to wire X (current into the plane of the paper)